1.3.1 ▶ ▶ ▶ Conservative Forces

This short section shows that defining a conservative force as a derivative of a scalar potential and a force which has a circulation of zero are equivalent.

The total work done on a particle going along a path p is

$$W = \int_p \mathbf{F} \cdot d\mathbf{r} \tag{1.33}$$

One definition of a conservative force is that it is able to be written as a derivative[5] of a scalar potential, i.e.

$$\mathbf{F} = -\frac{dU}{d\mathbf{r}} \tag{1.34}$$

Taking the dot product in the integral for work yields

$$W = \int_p \mathbf{F} \cdot d\mathbf{x} = \int_p F_x dx \tag{1.35}$$

Lastly, substituting our definition of a conservative force, we have

$$W = \int_p -\frac{dU}{dx} dx = \int_p -dU \tag{1.36}$$

Taking this integral over a closed path gives zero, since the starting and ending points are the same.

$$W = \oint_p -dU = 0. \tag{1.37}$$

Eq. 1.35 the other definition of a conervative force, and thus the two definitions are equivalent ★

[5] More rigorously, gradient